

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30	U04		MODULAR DBB	K15AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN
40	U04		MODULAR DBB	K15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
50	U04		MODULAR DBB	M15AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
60	U04		MODULAR DBB	M15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.
70	U04	VAW	MODULAR DBB	M15EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.
80										FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
80		VB2					***** BALL VALVES *****				
90										SB6	USE JACKSCREWS FOR PADDLE
100	U04	817	BALL	A15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0		BLANKS/SPACERS AND FLANGE SPREADERS
110	U04	817	BALL	A15CC05-K1	3	3	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0		FOR SPECTACLE BLINDS.
120	U04	817	BALL	A15CC07	4	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL
130	U04	817	BALL	A15CC07	10	12	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH
140	U04		BALL	B15AC01	20	20	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0		CHEMICAL SEALS.
150	U04	VA5	BALL	B15AC05	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
160	U04	817	BALL	B15CC05-K1	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB LO	0	0		VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
170	U04	817	BALL	B15CC07	3	3	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB LO	0	0		
180	U04	817	BALL	B15CC07	4	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0	VAW	
190	U04	817	BALL	B15CC07	10	12	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0		
200							BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0		
200		VQ2					***** THROUGH CONDUIT GATE VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
210										VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
210	U04	817	GATE	W15CC70	3	4	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
220	U04	817	GATE	W15CC71	20	20	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
230	U04	817	GATE	W15CC71	6	12	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
240										VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
240							***** GLOBE VALVES *****				
250											
250	U04		GLOBE	L15AC48	0.75	1.5	GLOBE 1500# RF LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0		
260			GLOBE	L15AC41	2	3	GLOBE 1500# RF LTCS A352 LCC INCONEL OS&Y HW	0	0		
270	U04	817	GLOBE	L15CC42	4	4	GLOBE 1500# CF LTCS A352 LCC INCONEL OS&Y GO	0	0		
280											
280							***** CHECK VALVES *****				
290											
290	U04	VC1	CHECK	C15AC50	0.75	1.5	CHECK PISTON 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM	0	0		
300	U04		CHECK	C15AC45	2	2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0		
310	U04	817	CHECK	C15CC52	4	4	CHECK DUAL DBL FLG 1500# CF LTCS A352 LCC INCONEL TRIM	0	0		
320	U04	817	CHECK	C15CC52	10	12	CHECK DUAL DBL FLG 1500# CF LTCS A352 LCC INCONEL TRIM	0	0		
330	U04	817	CHECK	C15CC52	20	20	CHECK DUAL DBL FLG 1500# CF LTCS A352 LCC INCONEL TRIM	0	0		
340											
340							***** PIPE *****				
350											
350			PIPE	5358511	0.75	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
360		812	PIPE	5358513	2	4	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
370			PIPE	5358525	10	10	PIPE WT 33.0MM SMLS LTCS A333 GR.6 BE	0	0		
380			PIPE	5358526	12	12	PIPE WT 37.0MM SMLS LTCS A333 GR.6 BE	0	0		
390			PIPE	5358527	20	20	PIPE WT 54.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		
400											
400							***** FITTINGS *****				
410											
410			ELBOW	5392870	0.75	3	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
420			ELBOW	5392800	10	10	ELL BW 90 DEG LR WT 33.0MM LTCS A420 WPL6	0	0		
430			ELBOW	5392801	12	12	ELL BW 90 DEG LR WT 37.0MM LTCS A420 WPL6	0	0		
440			ELBOW	5392809	20	20	ELL 90 DEG LR WT 54.0MM LTCS A420 WPL6-W	0	0		
450			ELBOW	5392876	0.75	3	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
460			ELBOW	5392802	10	10	ELL BW 45 DEG LR WT 33.0MM LTCS A420 WPL6	0	0		
470			ELBOW	5392803	12	12	ELL BW 45 DEG LR WT 37.0MM LTCS A420 WPL6	0	0		
480			ELBOW	5392810	20	20	ELL 45 DEG LR WT 54.0MM LTCS A420 WPL6-W	0	0		
490											
490			TEE	5392879	0.75	3	TEE SCH XXS LTCS A420 WPL6	0	0		
500			TEE	5392804	10	10	TEE BW WT 33.0MM LTCS A420 WPL6	0	0		
510			TEE	5392805	12	12	TEE BW WT 37.0MM LTCS A420 WPL6	0	0		
520			TEE	5392811	20	20	TEE BW WT 54.0MM LTCS A420 WPL6-W	0	0		
530			FLANGE	5392877	0.75	2	TEE RED SCH XXS LTCS A420 WPL6	1	4		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 126°C	12/16/2024	BY OZAC	U04	Page 1 of 3	1500K1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
540			FLANGE	5392806	10	10	TEE RED WT 37.0MM/33.0MM LTCS A420 WPL6	12	12	
550			FLANGE	5392812	10	10	TEE RED WT 54.0MM/ 33.0MM LTCS A420 WPL6-W	20	20	
560			FLANGE	5392813	12	12	TEE RED WT 54.0MM/ 37.0MM LTCS A420 WPL6-W	20	20	
570										
570	U04		CAP	5392880	0.75	3	CAP BW SCH XXS LTCS A420 WPL6	0	0	
580			CAP	5392631	10	10	CAP WT 33.0MM LTCS A420 WPL6	0	0	
590			CAP	5392808	12	12	CAP WT 37.0MM LTCS A420 WPL6	0	0	
600			CAP	5392814	20	20	CAP WT 54.0MM LTCS A420 WPL6	0	0	
610										
610	U04		RED CONC	5392881	0.75	3	REDUCER CONC SCH XXS LTCS A420 WPL6	2	4	
620			RED CONC	5392473	12	12	REDUCER CONC WT 54.0MM/37.0MM LTCS A420 WPL6-W	20	20	
630	U04		RED ECC	5392882	0.75	3	REDUCER ECC SCH XXS LTCS A420 WPL6	2	4	
640			RED ECC	5392522	12	12	REDUCER ECC WT 54.0MM/37.0MM LTCS A420 WPL6-W	20	20	
650										
650							***** OLETS *****			
660										
660	U04		OLETS	5504164	0.75	1.5	NIPOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	2	4	
670	U04		OLETS	5504149	0.75	2	NIPOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2 CL.1	10	10	
680	U04		OLETS	5504153	0.75	2	NIPOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2 CL.1	12	12	
690	U04		OLETS	5504135	0.75	2	NIPOFLANGE 1500# RF WT 54.0MM/XXS LTCS A350 LF2 CL.1	20	20	
700	U04	UJ2	OLETS	5504169	1.5	1.5	NIPOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	4	4	
710	U04	UJ2	OLETS	5504190	1.5	1.5	NIPOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2 CL.1	10	10	
720	U04	UJ2	OLETS	5504191	1.5	1.5	NIPOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2 CL.1	12	12	
730	U04	UJ2	OLETS	5504192	1.5	1.5	NIPOFLANGE 1500# RF WT 54.0MM/XXS LTCS A350 LF2 CL.1	20	20	
740										
740	U04		OLETS	5505197	0.75	1	WELDOLET SCH XXS LTCS A350 LF2 CL.1	2	2	
750	U04		OLETS	5505380	0.75	1	WELDOLET WT 37.0MM/SCH XXS LTCS A350 LF2 CL.1	12	12	
760	U04		OLETS	5505381	2	3	WELDOLET WT 54.0MM / SCH XXS LTCS A350 LF2 CL.1	20	20	
770										
770	U04	UJ2	ELBOFLG	5516077	1.5	1.5	ELBOFLANGE 1500# RF XXS LTCS A350 LF2 CL.1	4	4	
780	U04	UJ2	ELBOFLG	5516069	1.5	1.5	ELBOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2 CL.1	10	10	
790	U04	UJ2	ELBOFLG	5516070	1.5	1.5	ELBOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2 CL.1	12	12	
800	U04	UJ2	ELBOFLG	5516071	1.5	1.5	ELBOFLANGE 1500# RF WT 54.0MM/SCH XXS LTCS A350 LF2 CL.1	20	20	
810										
810		816					***** FLANGES *****			
820										
820	U04	FA8	FLANGE	5599704	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
830										
830			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
840			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
850										
850	U04		FLANGE	5557505	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
860	U04		FLANGE	5557505	3	4	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
870	U04		FLANGE	5557514	10	10	FLG WN 1500# RF LTCS A350 LF2 CL.1 WT 33.0MM BORE	0	0	
880	U04		FLANGE	5557515	12	12	FLG WN 1500# RF LTCS A350 LF2 CL.1 WT 37.0MM BORE	0	0	
890	U04		FLANGE	5557516	20	20	FLG WN 1500# RF LTCS A350 LF2 CL.1 WT 54.0MM BORE	0	0	
900	U04		FLANGE	5557706	2	2	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XXS	0	0	
910	U04	817	FLANGE	5619984	3	4	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 SCH XXS LINE CLASS 1500K1	0	0	
920	U04	817	FLANGE	5619846	10	10	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 WT 33.0MM	0	0	
930	U04	817	FLANGE	5619847	12	12	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 WT 37.0MM	0	0	
940	U04	817	FLANGE	5619848	20	20	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 WT 54.0MM	0	0	
950	U04		BLIND	5615616	0.5	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
960	U04		BLIND	5615616	3	20	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
970	U04		BLIND	5615614	0.75	20	FLG BLIND 1500# RTJ LTCS A350 LF2 CL.1	0	0	
980	U04	817	BLIND	5619786	3	4	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 CL.1 LINE CLASS 1500K1	0	0	
990	U04	817	BLIND	5619786	10	12	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 CL.1 LINE CLASS 1500K1	0	0	
1000	U04	817	BLIND	5619786	20	20	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 CL.1 LINE CLASS 1500K1	0	0	
1010										
1010							***** PIPE NIPPLES *****			
1020										
1020		812		0			BBE NIPPLES - USE PIPE CUT TO REQUIRED LENGTH			
1030										
1030							***** GASKETS *****			
1040										
1040			GASKET	5672251	0.5	20	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1050	U04	817	GASKET	5673481	3	20	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500K1	0	0	
1060			GASKET	5674460	2	2	GASKET 2500# ALLOY INC 625 OCTAGONAL RING	0	0	
1070										
1070							***** BOLTING *****			
1080										
1080	U04		BOLTING	5675770	0.5	14	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1090	U04	BA4	BOLTING	5675770	16	18	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1100	U04	BA4	BOLTING	5675776	20	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	

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1110	U04	BA1	BOLTING	5675776	16	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
1120		SB6	BOLTING	5675675	10	20	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1130			SPEC BLIND	5749595	1	2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
1140			PADDLE	5750049	10	20	PADDLE BLANK 1500# RF LTCS A516-GR70	0	0	
1150			PADDLE	5750249	10	20	PADDLE SPACER 1500# RF LTCS A516-GR70	0	0	
1160	U04	817	PADDLE	5750283	10	20	PADDLE BLANK COMPACT 1500# RJ LTCS A516-GR70	0	0	
1170	U04	817	PADDLE	5750284	10	20	PADDLE SPACER COMPACT 1500# RJ LTCS A516-GR70	0	0	
1180										

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